

Neha Narula

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INTERESTS Distributed systems, security, cryptocurrencies, and digital money

EDUCATION **Massachusetts Institute of Technology** Cambridge, MA
Ph.D. in Computer Science June 2015

Advisors: Robert T. Morris and Eddie Kohler
Thesis: *Parallel Execution for Conflicting Transactions*

Massachusetts Institute of Technology Cambridge, MA
S.M. in Computer Science September 2010

Advisor: Robert T. Morris.
Thesis: *Distributed Query Execution on a Replicated and Partitioned Database*

Dartmouth College Hanover, NH
A.B. in Computer Science and A.B. in Mathematics June 2003

Advisor: Prasad Jayanti
Thesis: *Eliminating Complex Synchronization Instructions in the Contention-Free Case for Mutual Exclusion Algorithms*

RESEARCH **MIT Media Lab** Cambridge, MA
EXPERIENCE *Director, Digital Currency Initiative* May 2016 – present

Director of the Digital Currency Initiative at the MIT Media Lab. Leading a team of 10 including research scientists, developers, and operational staff. Activities include research, writing software, teaching classes, advising undergraduates and masters students, and fundraising. Select projects:

Stablecoins. We research and investigate financial and technology risks and opportunities for stablecoins.

Central bank digital currency. We do technology research to understand how to safely design central bank digital currency and solve challenges including scalability, enabling offline access, and preserving privacy. We engaged in sponsored research collaborations with the Bill and Melinda Gates Foundation, Federal Reserve Bank of Boston, Bundesbank, Bank of Canada, Bank of England, World Bank, and Bank for International Settlements.

Cryptocurrency security. We found a vulnerability in the Curl-P hash function used in the cryptocurrency IOTA. I wrote the code to efficiently find collisions and generate conflicting attack transactions. Based on this and another vulnerability a DCI developer found in Bitcoin Cash, we established a cryptocurrency security initiative to explore the question of whether decentralized networks can be secure at scale and disseminate best practices on cryptocurrency security and vulnerability disclosure. We conducted research on the economic security of proof-of-work, which secures trillions of dollars of cryptocurrency. Our work expands the space of strategies to secure proof-of-work and implements monitoring tools to detect illicit miner activity.

zkLedger. zkLedger is a distributed ledger which provides transaction privacy and provably-correct, third-party auditing. zkLedger hides the participants and amounts in transactions, but the transactions can still be publicly verified to show that financial invariants are maintained. By using non-interactive zero-knowledge proofs, zkLedger allows a third party to query the participants

to analyze the contents of the ledger, without revealing individual transactions. We designed, implemented, evaluated, and released zkLedger as an open source project.

Supervised work. Other work at the DCI includes developing and maintaining Bitcoin Core, the primary software used in the Bitcoin network.

MIT CSAIL

Research Assistant in Parallel and Distributed Operating Systems

Cambridge, MA

January 2008 – May 2015

Doppel. I created Doppel, an in-memory multi-core transactional database designed to improve performance on workloads with many conflicting transactions. We developed a new technique called phase reconciliation; we take advantage of commutativity and executing transactions in explicit phases in order to increase concurrency. Doppel provides a dramatic performance improvement over existing concurrency control algorithms (3-30×) on highly conflicting workloads.

Dixie. I wrote Dixie, a SQL query planner, optimizer, and executor which issues SQL queries written for one database over a database sharded and replicated over multiple servers. Dixie focuses on increasing inter-query parallel speedup and throughput by using table replicas to involve fewer servers in each query, and simplifies the process of moving an application from a single database to a sharded database.

INDUSTRY
EXPERIENCE

Block

Member, Board of Directors

July 2023 – present

I also serve on the Audit and Risk Committee and the Nominating and Corporate Governance Committee.

FINRA

Member, FinTech Industry Committee

June 2023 – present

Federal Reserve Bank of New York

Member, Innovation Advisory Council

March 2022 – present

Paypal

Member, Blockchain and Digital Currencies Advisory Council

February 2022 – April 2023

News.me/Digg

Data Scientist

New York, NY

June 2012 – August 2012

Member of the five-person engineering team which launched the new Digg.com in six weeks.

Designed and implemented a system for analyzing shared content on Twitter and Facebook, and using these and other signals generated trending, new, and breaking news.

Google, Inc.

Senior Software Engineer

Mountain View, CA

July 2003 – January 2011

Designed and developed a Linux security sandbox for untrusted code running in the Native Client framework. Helped launch the research prototype of Native Client.

Designed and developed a highly available, distributed storage and serving system for large binary objects with five other engineers. Launched and maintained the system while supporting several production applications and serving gigabits of traffic per second.

Launched Froogle, Google's shopping website, into Germany and France.

PUBLICATIONS

Lovejoy, J., Virza, M., Fields, C., Karwaski, K., Brownworth, A. and **Narula, N.** *Hamilton: A High Performance Transaction Processor for Central Bank Digital Currencies*. In Proceedings of the 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI). Boston, MA, 2023.

Su, L., Liu, Q.C. and **Narula, N.** *The Power of Random Symmetry-Breaking in Nakamoto Consensus*. In Proceedings of the 35th International Symposium on Distributed Computing, 2021.

Park, S., Specter, M., **Narula, N.** and Rivest, R.L. *Going from bad to worse: from internet voting to blockchain voting*. In Journal of Cybersecurity, 2021.

Heilman, E., **Narula, N.**, Tanzer, G., Lovejoy, J., Colavita, M., Virza, M. and Dryja, T. *Cryptanalysis of curl-p and other attacks on the IOTA cryptocurrency*. In IACR Transactions on Symmetric Cryptology, 2020. Invited to present at Blackhat and Real World Crypto.

Böhme, R., Eckey, L., Moore, T., **Narula, N.**, Ruffing, T. and A. Zohar. *Responsible Vulnerability Disclosure in Cryptocurrencies*. In Communications of the ACM. 2020.

Narula, N., Vasquez, W. and M. Virza. *zkLedger: Privacy-Preserving Auditing for Distributed Ledgers*. In Proceedings of the 11th USENIX Symposium on Networked Systems Design and Implementation (NSDI). Renton, WA, 2018.

Narula, N., Cutler, C., Kohler, E. and R. Morris. *Phase Reconciliation for Contended In-memory Transactions*. In Proceedings of the 11th USENIX Symposium on Operating Systems Design and Implementation (OSDI). Broomfield, Colorado, 2014.

Kate, B., Kohler, E., Kester, M., **Narula, N.**, Mao, Y. and R. Morris. *Easy Freshness with Pequod Cache Joins*. In Proceedings of the 7th USENIX Symposium on Networked Systems Design and Implementation (NSDI). Seattle, Washington, 2014.

Narula, N. and R. Morris. *Executing Web Application Queries on a Partitioned Database*. In Proceedings of the USENIX Conference on Web Application Development (USENIX WebApps). Boston, Massachusetts, 2012.

Chandra, R., Kim, T., Shah, M., **Narula, N.** and N. Zeldovich. *Intrusion Recovery for Database-backed Web Applications*. In Proceedings of the ACM Symposium on Operating Systems Principles (SOSP). Cascais, Portugal, 2011.

Yee, B., Sehr, D., Dardyk, G., Chen, J.B., Muth, R., Ormandy, T., Oksaka, S., **Narula, N.** and N. Fullagar. *Native Client: A Sandbox for Portable, Untrusted x86 Native Code*. In the IEEE Symposium on Security and Privacy (Oakland). Oakland, California, 2010. **Best Paper Award, Test of Time Award**

Yip, A., **Narula, N.**, Krohn, M. and R.T. Morris. *Privacy-Preserving Browser-Side Scripting with BFlow*. In Proceedings of the ACM European Conference on Computer Systems (EuroSys). Nuremberg, Germany, 2009.

Jayanti, P., Petrovic, S. and **N. Narula**. *Read/Write Based Fast-Path Transformation for FCFS Mutual Exclusion*. International Conference on Current Trends in Theory and Practice of Computer Science (SOFSEM). Berlin, 2005.

INVITED
PUBLICATIONS

Contributor, *Digital Currencies: The US, China, and the World at a Crossroads*. Eds. Darrell Duffie and Elizabeth Economy. Hoover Institution, Stanford University, March 2022.

Contributor, *Digital Currencies: Risk or Promise?* Cato Journal, 2021.

Casey, M., Crane, J., Gensler, G., Johnson, S. and **N. Narula**. *The Impact of Blockchain Technology on Finance: A Catalyst for Change*. ICMB, International Center for Monetary and Banking Studies, 2018.

POSTS,
ABSTRACTS, AND
REPORTS

Aronoff, D., Calabia, C., Brownworth, A., Samuel, A. and **N. Narula**. *The Hidden Plumbing of Stablecoins: Financial and Technological Risks in the GENIUS Act Era*. January 2026.

Toh, W. K., Maurer, M., Landriault, E., Samuel, A., Wang, L. and **N. Narula**. *Designing Payment Tokens For Safety, Integrity, Interoperability, and Usability*. May 2025.

Lovejoy, J., Brownworth, A., Virza, M. and **N. Narula**. *PARSEC: Executing Smart Contracts in Parallel*. October 2023.

George, N., Dryja, T. and **N. Narula**. *A Framework for Programmability in Digital Currency*. August 1, 2023.

Narula, N., Swartz, L. and Frizzo-Barker, J. *CBDC: Expanding Financial Inclusion or Deepening the Divide? Exploring Design Choices that Could Make a Difference*. January 12, 2023.

Auer, R., Frost, J., Lee, M., Martin, A., and **N. Narula**. *Why Central Bank Digital Currencies?* NY Fed Liberty Street Economics blog, December 1, 2021.

Liu, Q., Dryja, T. and **N. Narula**. *A Lower Bound for Byzantine Agreement and Consensus with Adaptive Adversaries using VDFs*.

Cline, D., Dryja, T. and **N. Narula**. *Clockwork: An Exchange Protocol for Proofs of Non Front-Running*.

Moroz, D., Aronoff, D., Lovejoy, J., **Narula, N.** and D. Parkes. *Double-Spend Counter-Attacks: Threat of Retaliation in Proof-of-Work Systems*.

Narula, N. and C. Fields. *Reducing the Risk of Catastrophic Cryptocurrency Bugs*. Medium post, August 9, 2018.

Aspegren, H., Glasbergen, G., Weber, M. and **N. Narula**. *b_verify: Scalable Non-Equivocation for Managing Public Data*.

Barabas, C., **Narula, N.** and E. Zuckerman. *Back to the Future: The Decentralized Web*. Report, 2017.

N. Narula. *A Multi-core Database is not a Distributed System*. In the Conference on Innovative Data Systems Research (CIDR). Asilomar, California, 2015.

Narula, N. and R. Morris. *Designing a Toolkit for Distributed Storage in Web Applications*. Poster at the Symposium on Operating Systems Principles (SOSP). Big Sky, Montana, 2009.

SERVICE

Board member, Waye
Program Committee, Financial Cryptography

2025-present
2025

	Program Committee, Advances in Fintech Technology	2024
	Program Committee, NSDI	2023
	Invited academic expert, World Economic Forum Annual Meeting	2023
	Co-chair, ACM Advances in Fintech Technology	2022
	Program Committee, OSDI	2022
	Program Committee, Financial Cryptography	2021
	Program Committee, ACM Advances in Fintech Technology	2021
	Program Committee, Financial Cryptography	2020
	Invited academic expert, World Economic Forum Annual Meeting	2020
	Program Committee, IEEE Security and Privacy	2020
	Program Committee, Stanford Blockchain Conference	2020
	Program Committee, ACM Symposium on Cloud Computing	2019
	Program Committee, EuroSys	2019
	External Reviewer, PODC	2019
	Member, World Economic Forum's Global Blockchain Council	2019-2020
	Co-editor-in-chief and cofounder, Journal of Cryptoeconomic Systems (MIT Press)	2019-2024
	Program Committee, Scaling Bitcoin	2016
	Program Chair, Scaling Bitcoin	2015
	Resident at Hacker School (now the Recurse Center)	2015
	MIT EECS Faculty Search Student Subcommittee	2015
	Leading MIT's distributed systems reading group	2014-2015
	Google Mentoring Committee	2006-2008
	Google Foundation Steering Committee	2003
STUDENTS ADVISED	Ayesha Ali, MEng CS, MIT (Instabase)	2023-2024
	Claire Bao, MEng CS, MIT (Jump Trading)	2023-2024
	Shwetark Patel, MEng CS, MIT (startup)	2021-2022
	James Lovejoy, MEng CS, MIT (Director of Engineering at the Boston Fed)	2019-2020
	Henry Aspegren, MEng CS, MIT (Google, Meta, OpenAI)	2017-2018
	Willy R. Vasquez, MEng CS, MIT (PhD UT Austin, Apple)	2016-2017
TEACHING	Cryptocurrency Design and Engineering (MIT MAS.S62)	Fall 2025
	MIT/GetSmarter online cryptocurrency course	
	Co-lead with Gary Gensler	Fall 2019
	Blockchain Lab (MIT 15.S68, 15.217)	
	Co-lecturer with Luis Barros, Gary Gensler, and Simon Johnson	Spring 2021
	Co-lecturer with Michael Casey, Gary Gensler, and Simon Johnson	Spring 2019, 2020
	Cryptocurrency Engineering and Design (MIT MAS.S62)	
	Co-lecturer with Tadge Dryja. Available on MIT Open Courseware.	Spring 2018
	Shared Public Ledgers: Cryptocurrencies, Blockchains, and Other Marvels (MIT 6.892)	
	Co-lecturer with Silvio Micali	Spring 2017
	Distributed Systems (MIT 6.824)	
	Teaching assistant, guest lecturer	Spring 2013
	Computer Systems Engineering (MIT 6.033)	
	Teaching assistant	Spring 2011

SELECT MEDIA	TED.com. <i>The future of money</i> (3M+ views)	
	MIT Technology Review. <i>The MIT researcher who helps senators understand digital currencies</i>	
	CBS 60 minutes. <i>Bitcoin's Wild Ride</i>	
	Wall Street Journal. <i>Does the U.S. Need a National Digital Currency?</i>	
	The New Yorker Live. <i>How Memes Become Money</i>	
	Amanpour & Co. <i>Currency Futurist Neha Narula Debunks Cryptocurrency</i>	
	Wired.com. <i>The Blockchain: Boon for Bankers or Tool for Tyrants?</i>	
	Techcrunch.com. <i>Cryptocurrency Insecurity: IOTA, BCash and Too Many More</i>	
	Motherboard.com. <i>A \$5 Billion Cryptocurrency Has Enraged Cryptographers</i>	
	CNBC. <i>Digital Currency Could Change How We Deal with Money</i>	
	PBS Newshour. <i>The How and Why of Buying Bitcoin</i>	
	Wired.com. <i>Decentralized Social Networks Sound Great. Too Bad They'll Never Work</i>	
	Harvard Business Review. <i>The Blockchain Will Do to the Financial System What the Internet Did to Media</i>	
	Wired.com. <i>MIT Computer Scientists Demonstrate the Hard Way That Gender Still Matters</i>	
	Reddit.com. <i>We're 3 Female Computer Scientists from MIT. Ask us anything!</i>	
HONORS AND AWARDS	Rockefeller Foundation Bellagio Center Residency	2025
	IEEE Symposium on Security and Privacy Test of Time Award	2021
	IMSA Alumni Trailblazer Award	2021
	WIRED 25 Leaders Shaping the Next 25 Years of Technology	2019
	Academy of Achievement Delegate	2019
	Thinkers50 Radar list	2018
	Fortune's The Ledger 40 under 40 list	2018
	IEEE Symposium on Security and Privacy Best Paper Award	2010
	Eben Tisdale Fellowship (declined)	2009
	NSF Graduate Research Fellowship	2007
	High Honors in Computer Science	2003
SELECT INVITED TALKS	The Hidden Plumbing of Stablecoins	
	New York Fed Innovation Conference, New York, NY.	June 2026
	Boston Fed Fintech Hub seminar, virtual.	May 2026
	Bank of England seminar, virtual.	March 2026
	System requirements and design choices for private, scalable digital cash	
	Bank for International Settlements, Basel, Switzerland.	August 2025
	Advances in Fintech (keynote), Vienna, Austria.	September 2024
	Can Bitcoin Self-Custody Scale to a Billion Users?	
	BITCOIN 2025, Las Vegas, NV.	May 2025
	MIT Bitcoin Expo, Cambridge, MA.	April 2025
	Plan B Forum, El Salvador.	January 2025
	Economic Security of Proof-of-Work	
	MIT Bitcoin Expo, Cambridge, MA.	April 2024
	Chaincode, New York, NY.	July 2019
	Central Bank Digital Currency: Risks and Opportunities	
	Hoover Institute, virtual.	July 2021
	Digitizing the Dollar	
	US congressional testimony before the House Task Force on Financial Technology	June 2021

Building A Stronger Financial System: Opportunities of a Central Bank Digital Currency	
US congressional testimony before the Senate Economic Policy Subcommittee	June 2021
Redesigning Digital Money: What Can We Learn from a Decade of Cryptocurrencies?	
Bank of Canada, Ottawa, Canada.	October 2019
The Architecture of Crypto Innovation	
a16z Crypto Regulatory Summit, San Francisco, CA.	May 2019
Preventing Catastrophic Cryptocurrency Attacks	
MIT Bitcoin Expo, Cambridge, MA.	March 2019
Financial Cryptography (keynote), St. Kitts.	February 2019
A Tangled Curl: How We Forged Signatures in IOTA	
Real World Crypto, San Jose, CA.	January 2019
Blackhat, Las Vegas, NV.	August 2018
zkLedger: Privacy-Preserving Auditing for Distributed Ledgers	
NBER Cryptocurrencies Workshop, Cambridge, MA.	May 2019
Fintech@CSAIL Annual Meeting, Cambridge, MA.	September 2018
PODC Blockchain Workshop, Egham, UK.	July 2018
Microsoft Research, Redmond, WA.	April 2018
NSDI, Renton, WA.	April 2018
MIT Bitcoin Expo, Cambridge, MA.	March 2018
Technion Summer School on Cyber and Security, Haifa, Israel.	September 2017
21st Century Alchemy: Creating the Internet of Value	
Depository Trust and Clearing Corporation, New York, NY.	April 2019
Goldman Sachs, New York, NY.	May 2018
The Future of Money	
SXSW, Austin, TX.	March 2018
EmTech China, Beijing, China.	January 2018
Banco Central de Chile, Santiago, Chile.	December 2017
TED@BCG, Paris, France (3M views).	May 2016
Trading Simplicity for Performance When Designing Distributed Systems	
Mesosphere, San Francisco, CA.	October 2015
MesosCon (keynote), Seattle, WA.	August 2015
Splitting and Replicating Data for Fast Transactions: Don't Give Up on Serializability Just Yet	
OREDEV, Malmo, Sweden.	November 2015
CRAFT, Budapest, Hungary.	April 2015
GOTO Chicago, Chicago, IL.	April 2015
Papers We Love: The Scalable Commutativity Rule	
Papers We Love, New York, NY.	April 2015
A Multi-core Database Is Not a Distributed System	
CIDR short talk, Asilomar, CA.	January 2015
Phase Reconciliation for Contended In-Memory Transactions	

RICON, Las Vegas, NV.	October 2014
OSDI, Broomfield, CO.	October 2014
MIT Industry Affiliate Program Cloud Workshop, Cambridge, MA.	September 2014
 Consensus and Consistency: Why Should I Care?	
Berlin Buzzwords, Berlin, Germany.	May 2014
 The Good, the Bad, and the Ugly (of Caching)	
All Your Base (keynote), Oxford, UK.	October 2013
 Smarter Caching With Pequod	
RICON East, New York, NY.	May 2013
 Executing Queries on a Sharded Database	
Future of Webapps, Las Vegas, NV.	October 2013
Future Insights Live, London, UK.	May 2013
Strangeloop, St. Louis, MO.	September 2012
Hacker School, New York, NY.	August 2012
USENIX Webapps, Boston, MA.	June 2012